Edward Sinclair

Work Portfolio

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Drone-based Robotic Chainsaw Tool

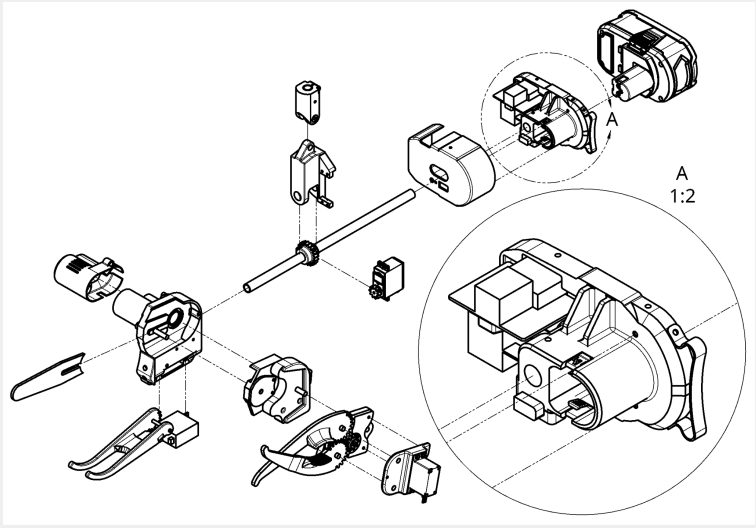
Master's Project - ⅓ scope





Purpose

- Develop a lightweight robotic pruning tool for drone-based operation in hazardous and difficult-to-reach locations



Engineering Highlights

- Lightweight 3D-printed PLA and carbon fibre tube structure
- Custom housing designed around reverse-engineered chainsaw internals
- ESP32-based controller for wireless communication, tool actuation, and automated procedures
- Four-servo mechanism for tool orientation, branch feeding, gripping, and emergency release
- Custom Ryobi battery interface

Outcome

- Successfully demonstrated reliable in-flight branch pruning to a Radiata pine breeding company and on TVNZ's *Seven Sharp*
- The design's success inspired a fourth-year Mechanical Engineering Honours project

Automated RCD Coil Testbench

Basis NZ Ltd

Purpose

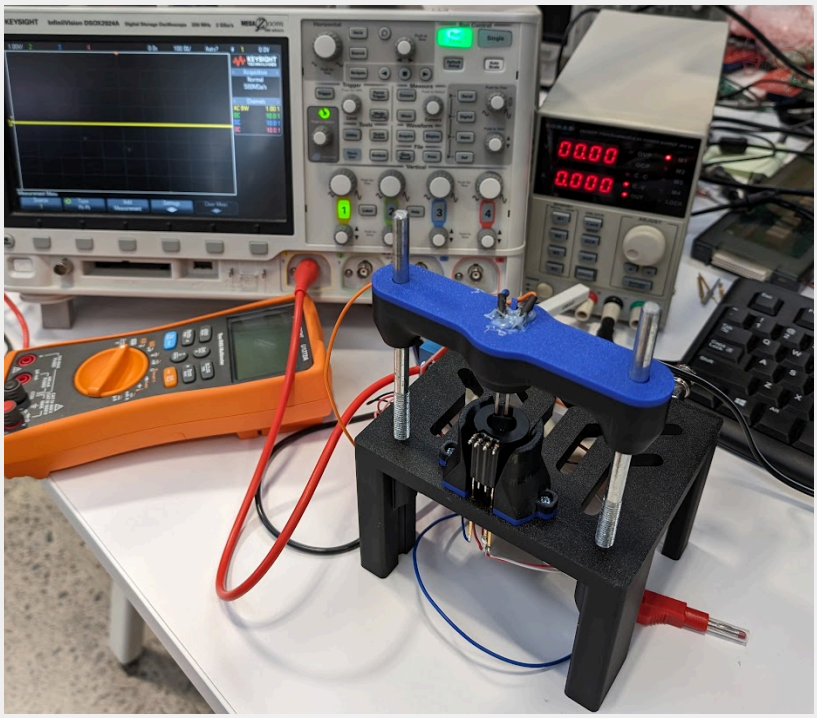
- Rapidly validate RCD coil performance to identify faulty production units

Engineering Highlights

- 3D printed PLA main structure
- Pogo-pin PCB interface for rapid control board replacement
- Custom linear guide assembly and internal power rail system
- Integrated relay array for automated power and signal routing
- Designed for repeatable testing of all 6 coil contact points

Outcome

- Designed and built in < 2 weeks
- Tested up to 16 coils per minute
- Used as a reference for scaling factory validation by an overseas engineering team



Telescope Camera Focus and Alignment Assembly

Engineering Honours Research Project - ⅓ scope

Purpose

- Develop an externally adjustable camera alignment and focus assembly for internal telescope installation

Engineering Highlights

- Aluminium structure comprising machined blocks, tubing, and waterjet-cut plates
- Compact 3D-printed focus mechanism with linear sliders, lead screw, and NEMA 17 stepper motor
- Integrated limit switches for travel protection
- Designed to allow camera alignment and focus adjustment after installation

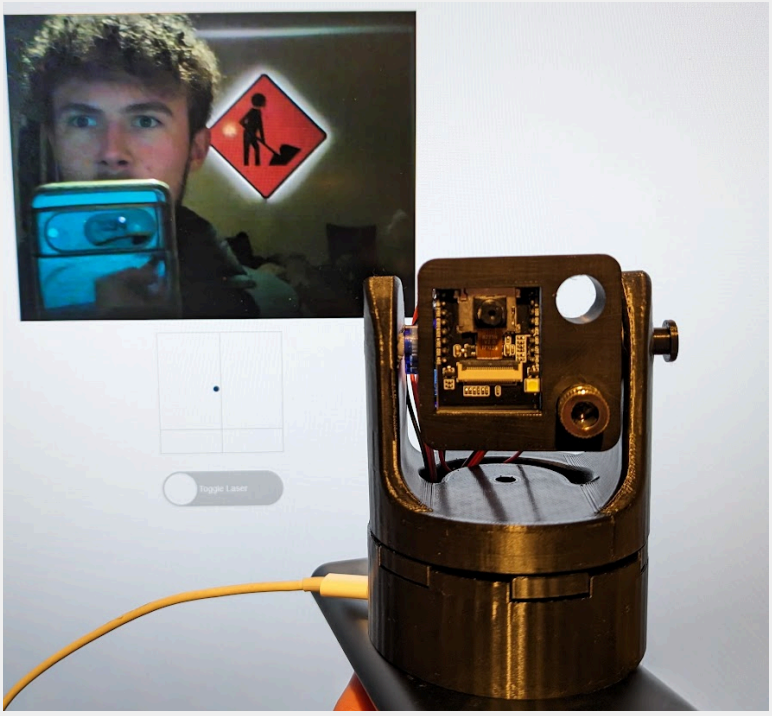
Outcome

- Enabled external collimation of the telescope camera with the primary mirror



Remote/Autonomous camera+laser module

Computer Vision Project



Purpose

- Develop a remotely operated pan-tilt camera system with integrated laser pointer

Engineering Highlights

- Custom 3D-printed enclosure
- Two-servo pan and tilt mechanism
- ESP32-CAM module providing wireless video streaming and remote control
- Designed using low-cost, off-the-shelf components with a total hardware cost under $15

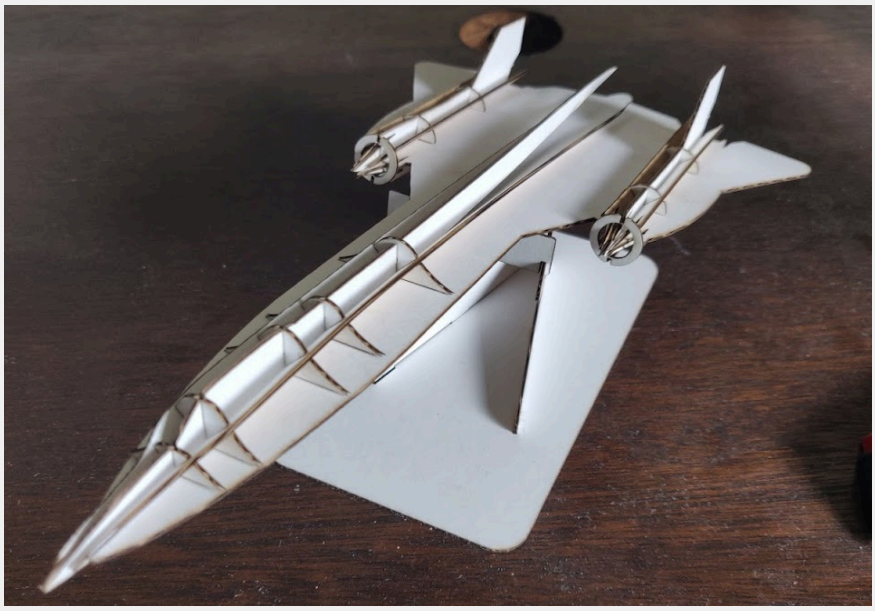
Outcome

- Fully functional proof-of-concept demonstrating remote video streaming and laser positioning

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Laser-cut SR-71 Blackbird Model

Personal Project



Purpose

- Fun design project to familiarise myself with laser cutting procedures

Engineering Highlights

- Designed entirely from reference images
- Self-interlocking tab-and-slot cardboard construction
- Optimised for assembly without adhesives

Outcome

- Successfully fabricated and assembled as designed